Making compounds

11a

When atoms lose or gain electrons, they become positive or negative ions. Positive ions join with negative ions to form compounds. In this activity positive ions have a small square sticking out, while negative ions have a small square missing.

hydrogen H^+	hydrogen H^+	chlorine Cl^-	chlorine Cl^-
hydrogen H^+	sodium Na^+	chlorine Cl^-	hydroxide OH^-
sodium Na^+	sodium Na^+	hydroxide OH^-	hydroxide OH^-
magnesium Mg^{2+}	magnesium Mg^{2+}	oxygen O^{2-}	oxygen O^{2-}
calcium Ca^{2+}	calcium Ca^{2+}	oxygen O^{2-}	sulfate SO_4^{2-}
copper Cu^{2+}	zinc Zn^{2+}	sulfate SO_4^{2-}	sulfate SO_4^{2-}
iron Fe^{3+}	iron Fe^{3+}	carbonate CO_3^{2-}	nitrate NO_3^-
			nitrate NO_3^-

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Making compounds 11b

1 Cut out the positive and negative ions and arrange them to make some compounds. A compound is a large rectangle made up of positive and negative ions, with the small squares sticking out to fit into the missing squares.

Naming compounds

When the formula of a compound is written down, the positive ion is written first and then the negative ion. Subscripts indicate the number of ions involved. For example, HHO becomes H_2O_1. Note that '1' does not need to be written down, so the correct formula is H_2O.

2 Make four compounds and record their formulas by counting the number of ions used.

When writing the name of the compound, only the second part of the name requires attention. The first part of the name is the name of the positive ion (which is often a metal). For the second part, the name of an element changes so that it ends in 'ide'. For example, oxygen becomes oxide. When negative ions ending in 'ate' join with metals, their name stays the same. When they join with hydrogen, the whole name changes. For example, when hydrogen and carbonate join, the new compound is called carbonic acid.

3 Construct six compounds and record their formulas and names in the table below.

Compound	Formula	Name
1		
2		
3		
4		
5		
6		

4 Make compounds out of the ions specified in the table below. Then write their formulas and names in the table.

Positive ion	Negative ion	Formula	Name
hydrogen	chloride		
hydrogen	sulfate		
hydrogen	hydroxide		
magnesium	sulfate		
magnesium	oxide		
magnesium	hydroxide		
iron III	chloride		
iron III	hydroxide		
iron III	sulfate		
		Fe_2O_3	
		$CuSO_4$	
		$Cu(NO_3)_2$	
		Na_2O	

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